Case Study Part 1: Menopausal Hormone Therapy

Overview

- What is menopause?
- What is Menopausal Hormone Therapy?
- Status of research prior to Women's Health Initiative
- The Women's Health Initiative
- Risk as an estimand
- Recent observational studies & next steps

What is menopause?

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- Officially in menopause when you have gone 12 months without menstruating
    - The time leading up to menopause is called peri-menopause
    - After the 12-month mark, you are "menopausal" or postmenopausal

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- diminished sleep quality
- insomnia
- irritability
- depression
- mood swings
- difficulty concentrating
- reduced quality of life
- aching joints
- bloating
- breast tenderness
- bladder/urinary discomfort
- more frequent urinary tract infections
- vaginal dryness, discharge, and bleeding
- loss of sexual desire, painful intercourse
- headaches
- thinning hair
- heart palpitations

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- 1960s: "Feminine Forever": youth, beauty, full sex life for menopausal women who take estrogen
- 1970s: Criticism of "paternalistic" medical establishment labeling menopause as a "deficiency disease" when it is a natural part of aging. Argue against using the term hormone "replacement"

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  - "...estrogen should be in the water... we thought it was like fluoride" -Hadine Joffe, Harvard Medical School professor of psychiatry

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- Between 1975 and 2000, of twenty published studies examining menopausal hormone therapy, 80% found no increased risk of breast cancer, 10% found an increased risk of breast cancer, 10% found a decreased risk of breast cancer (meta-analysis sponsored by Wyeth)

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- Study halted July 2002 (average follow-up 5.2 years)

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- What do we mean by increased risks?
  - In 1 year for 10,000 postmenopausal women taking estrogen+progesterone, 8 more will have invasive breast cancer, 7 more will have a heart attack, 8 more will have a stroke, 18 more will have blood clots, than a similar group of 10,000 women not taking the hormones (Dr. Rossouw, acting director WHI)

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  - contextualization of risks

WHI Results

	CEE + MPA Trial			
	No. (%) of Events[a]			
	CEE + MPA (n=8506)	Placebo (n=8102)	Difference/ 10 000 PY[b]	HR (95% CI)
Primary end points				
Coronary heart disease				
50-59 y	38 (0.23)	27 (0.17)	5	1.34 (0.82-2.19)
60-69 y	79 (0.37)	73 (0.37)	0	1.01 (0.73-1.39)
70-79 y	79 (0.82)	59 (0.63)	19	1.31 (0.93-1.84)
Invasive breast cancer				
50-59 y	55 (0.33)	42 (0.27)	6	1.21 (0.81-1.80)
60-69 y	97 (0.45)	75 (0.38)	8	1.20 (0.89-1.62)
70-79 y	54 (0.56)	38 (0.41)	15	1.37 (0.90-2.07)

WHI Results

Age	N CEE+MPA	CEE+MPA B.C.	N Placebo	Placebo B.C.
50-59	2837	55	2683	42
60-69	3854	97	3655	75
70-79	1815	54	1764	38

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  - Relate time that passes before an event occurs (e.g. breast cancer, stroke, death) to covariates
  - Among individuals without breast cancer, what is the probability they develop breast cancer in the next year? The paper then pools this over all years into a hazard ratio (HR)

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- In 2002, the WHI Data and Safety Monitoring Board calculated an HR of 1.26, a 26% increase in breast cancer risk under treatment
- The WHI and media reporting on the halt of the study have been criticized for failing to communicate the difference between absolute and relative risks ($\frac{8}{10000} \neq .26$)

Results from WHI

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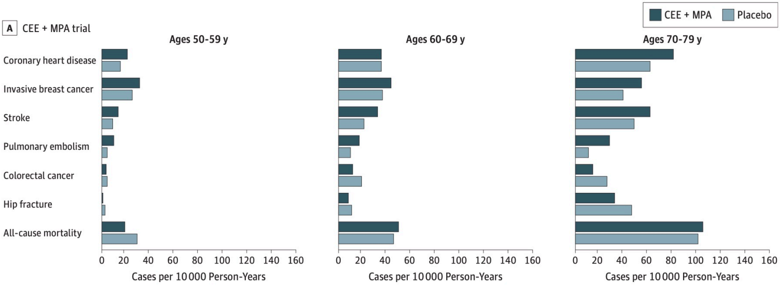
- Overall, 9 more cases of invasive breast cancer per 10,000 person-years

Figure 3. Absolute Risks of Health Outcomes by 10-Year Age Groups in the Women's Health Initiative Hormone Therapy Trials During the Intervention Phase



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  - Women who scored below cutpoints were given additional testing for dementia (follow ups continued regardless of dementia status)

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- CEE+MPA difference from placebo averaged 0.18 points (SE 0.10)
- Difference is "too small to have relevance in clinical practice"

Results from WHIMS

Table 2. Mean Modified Mini-Mental State Examination Scores by Time From WHI Randomization and Treatment Assignment

No. of Years Since Randomization	Estrogen + Progestin: No. of Patients	Estrogen + Progestin: Mean (SD)	Placebo: No. of Patients	Placebo: Mean (SD)	Difference Between Treatments, Mean (95% CI)
0	2132	95.50 (4.21)	2215	95.63 (3.87)	−0.13 (−0.37 to 0.11)
1	2102	95.98 (4.10)	2188	96.20 (3.70)	−0.22 (−0.46 to 0.01)
2	2022	96.37 (3.92)	2095	96.50 (3.91)	−0.13 (−0.37 to 0.11)
3	2005	96.38 (4.36)	2083	96.74 (3.72)	−0.36 (−0.61 to −0.11)
4	1814	96.38 (4.96)	1875	96.90 (3.92)	−0.50 (−0.79 to −0.21)
5	833	96.71 (4.66)	901	96.87 (4.26)	−0.16 (−0.58 to 0.26)
6*	31	96.81 (2.39)	44	96.16 (8.63)	0.65 (−2.53 to 3.82)

Abbreviations: CI, confidence interval; WHI, Women's Health Initiative.
*Includes 9 women who were tested at 7 years after randomization.

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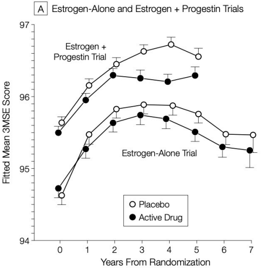
Figure 2. Fitted Mean Modified Mini-Mental State Examination Scores for Estrogen-Alone and Estrogen Plus Progestin Trials and Pooled Trials



No. of Women								
Estrogen + Progestin Trial								
Placebo	2213	2165	2078	2060	1853	822		
Estrogen + Progestin	2131	2088	2010	1992	1803	880		
Estrogen-Alone Trial								
Placebo	1421	1381	1313	1309	1191	1148	1088	209
Estrogen	1387	1355	1278	1261	1199	1121	1052	204

No. of Women						
Placebo	3634	3546	3391	3369	3044	1970
Any Hormone Treatment	3518	3443	3278	3253	3002	2001

WHI(MS) vs Observational Studies

Why are the results from the pre-WHI literature so different from the WHI(MS) results?

Recent Observational Studies

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- Recent observational studies look at different formulations/doses, administrations, durations of use, such as:
  - micronized bioidential progesterones versus synthetic progesterone (MPA)
  - pill/tablets, patches, and injections
  - durations of use: 1 year or less, 1-3 years, 3 to 6 years, longer than 6 years

Next Steps

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- Later: read about the effect of WHI/WHIMS on perceptions and prescriptions
- Later: analyze WHI- and WHIMS-based data with covariates
- Later: look closer at these recent observational studies with a focus on different levels of treatment